Fiscal Year:	FY 2012	**Task Last Updated:**	FY 04/05/2012
PI Name:	Azzam, Edouard Ph.D.		
Project Title:	The Role of Gap-Junction Communication and Oxidative Metabolism in the Biological Effects of Space Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	07103	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
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No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date extended until 12/31/2011, per C. Guidry/JSC (10/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	de Toledo , Sonia (UMDNJ-New Jersey Medical School) Harris, Andrew (New Jersey Medical School (UMDNJ)) Howell, Roger (University of Medicine and Dentistry of New Jersey) Pain, Debkumar (UMDNJ - New Jersey Medical School)		
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Task Description:	The objective of this project is to investigate biochemical and molecular aspects of two inter-related fundamental biological processes, namely gap-junction intercellular communication and oxidative metabolism, in determining human cellular responses to exposures of very low to moderate fluences of protons (of energies such as found in the space environment) or high charge (Z) and high energy (E) particles (1 GeV/nucleon 56Fe ions). Gap junctions are found between most cells in the body, and the molecular signaling they mediate is crucial for normal development, physiology and response to disease. By allowing direct intercellular transfer of cytoplasmic molecules, they provide a powerful pathway for direct molecular signaling between cells. On the other hand, oxidative metabolism is primarily regulated by mitochondria, which mediate respiration and ATP synthesis in eukaryotic cells. They also participate in numerous indispensable metabolic pathways. We hypothesize that exposure of human cells to high or low fluences of protons and/or HZE particles results in distinct molecular and biochemical changes not only in irradiated but also neighboring bystander cells. These events are modulated by oxidative metabolism and gap-junction coupling, which together induce persistent changes in mitochondria and in nuclear DNA in progeny of irradiated and affected bystander cells. In Aim 1, we propose to define the aspect(s) of junctional communication leading to proton- and high charge/high energy (HZE) particle-induced toxic effects. Specifically, we will examine the role of different connexin proteins in propagating death-inducing (growth delay, apoptosis) or pro-survival effects (proliferative capacity) among irradiated cells and also between irradiated and non-irradiated cells. Aim 2 is to identify specific connexin channel permeabilities that correlate with intercellular propagation of toxic or protective effects in cultures exposed to HZE particles or protons. An emphasis will be to determine the nature of molecules communicated among irradiated cells and between irradiated and non-irradiated cells. In Aim 3, we will evaluate mitochondrial function in cells exposed to low fluences of high energy charged particles. Specifically, we will examine mitochondrial protein import, mitochondrial protein synthesis, assembly of large protein complexes including respiratory complexes, and oxidative damage to mitochondrial proteins and DNA. We will also evaluate the persistence of oxidative effects in the progeny of exposed cells and correlate the observed results with the overall cellular redox environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed experiments relate to Risk No. 28, subsection 28b (molecular studies on carcinogenesis) in NASA's 'Roadmap'. In addition to revealing signaling pathways underlying non-targeted effects of ionizing radiation, the outcome of this proposal may provide insight into space radiation-induced degenerative diseases such as cognitive defects and cardiovascular disorders (Risks No. 29 and 30). It will provide a mechanistic basis for understanding exposures and risks to human from low fluences of high and low linear energy transfer radiations. By identifying the nature of molecules communicated among irradiated and neighboring non-irradiated cells, it may contribute to remediation strategies against harmful effects of space radiation, and is also relevant to heavy ion and proton radiotherapy.
Task Progress:	We have further extended our in vivo investigation of the role of intercellular communication and oxidative metabolism in the biological responses to high charge (Z) and high energy (E) HZE particles. We have exposed 12-week-old and 9-months-old male mice to low (5 cGy) or moderate (50 cGy) mean absorbed doses of 1000 MeV/amu iron ions or 500 MeV/amu titanium ions delivered to the whole body. In young mice, we have identified that regulation of the MAF1 protein, a downstream effector of the TOR (target of rapamycin) pathway that regulates various cellular functions in response to environmental cues, is highly sensitive to space particulate radiations. Maf1 has been implicated in processes related to protein biosynthesis and RNA maturation. It is essential for the activity of the enzyme RNA polymerase III that regulates cell growth and cell cycle progression. We have also performed global analyses of post-translational modification of proteins in organs of control and irradiated young mice. Specifically, we have analyzed S-nitrosylation of proteins from mice exposed to 5 or 50 cGy of energetic titanium ions. S-nitrosylation is emerging as a fundamental mechanism that modulates the function of several proteins engaged in many signaling pathways. Our results showed that S-nitrosylation of proteins in space irradiated rodents is organ and radiation dose dependent. We found that the S-nitrosylation level of brain proteins was significantly increased after whole body exposure to a mean absorbed dose of 5 cGy. Compared to control, 151 proteins were identified as newly S-nitrosylated 12 days after irradiation. A large fraction of the proteins (56%) were classified as enzymes. Pathway analysis showed that they are related to free radical scavenging, cellular assembly and organization. Analyses of kinetic changes in oxidative metabolism, immune responses and intercellular communication in organs of irradiated middle-age mice are in progress. We have also extended our in vitro studies in progeny of non-targeted bystander cells that were co-cultured with HZE-particle irradiated cells. We have analyzed several biological endpoints, including chromosomal damage, gene expression and cellular senescence. We have further confirmed that stressful effects propagated from irradiated to bystander cells persist in their progeny. We have shown that functional gap junction coupling between parental bystander and irradiated donor cells is a critical mediator of the observed stressful effects
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